



ARAVALI INTERNATIONAL SCHOOL
SESSION 2026-27
Grade- VII
SUBJECT - MATH

TOPIC- Constructions and Tillings

Name: _____ **Roll No.:** _____ **Teacher's Signature:** _____

Section A – Multiple Choice Questions

1. To construct the perpendicular bisector of a line segment AB, the arcs drawn from A and B should have:

- a) Different radii ☒ b) The same radius greater than half of AB
c) Radius less than half of AB d) No fixed radius

2. Which of the following angles can be constructed by first constructing a 60° angle and then bisecting it?

- a) 15° ☒ c) 30°
b) 45° d) 120°

3. If two rays of an angle are copied using equal arcs and equal distances, the new angle is:

- ☒ a) Twice the original angle c) Half the original angle
b) Equal to the original angle d) Always 90°

4. Which of the following regular polygons can tessellate a plane by itself without gaps or overlaps?

- a) Regular pentagon ☒ c) Regular hexagon
b) Regular heptagon d) Regular octagon

5. Which construction divides an angle into two equal parts?

- a) Perpendicular bisector ☒ c) Angle bisector
b) Parallel line d) Transversal

Section B – Assertion and Reasoning

Choose the correct option:

- a. Both Assertion (A) and Reason (R) are true, and R is the correct explanation of A.
b. Both A and R are true, but R is not the correct explanation of A.
c. A is true, but R is false. d. A is false, but R is true.
6. Assertion (A): A 30° angle can be constructed using a compass and ruler.

Reason (R): A 30° angle can be obtained by bisecting a 60° angle.

7. Assertion (A): A regular hexagon can tessellate a plane without gaps or overlaps.

Reason (R): The interior angle of a regular hexagon is 120° , and three such angles make 360° around a point.

Section C – Construction-Based Questions

8. Draw an angle $\angle ABC$. Using only a compass and ruler, construct a copy of $\angle ABC$ at a different point. Write the steps of construction.

9. Draw an angle of approximately 80° . Using a compass and ruler, construct its angle bisector. Write the steps of construction.

10. Draw a line segment $PQ = 6$ cm. Using a compass and ruler, construct its perpendicular bisector. Mark the point of intersection as M. Write the steps of construction.

11. Construct angles using compass and ruler.

a) 60° b) 30°

Show all construction arcs clearly and write the construction steps for either one.

Section D – Tessellations

12. A floor is to be covered completely using only one type of regular polygon, without leaving gaps or overlaps. Which of the following can not be used? Give a reason.

a) Equilateral triangle b) Square c) Regular hexagon d) Regular pentagon

13. A 5×7 grid has 35 unit squares. Can it be completely tiled using 2×1 rectangular tiles without cutting any tile? Explain your answer.

14. A regular hexagon is used to make a tessellation. What is the measure of each interior angle?

Section E – Case Study

15. A courtyard is $6 \text{ m} \times 4 \text{ m}$ and is to be covered with $2 \text{ m} \times 1 \text{ m}$ rectangular tiles.

a) How many tiles are needed to cover the courtyard completely?

b) Can the tiles be placed horizontally or vertically? Explain.

c) Draw a simple representation of one possible tessellation of the courtyard.